MATHEMATICS

University *of* NORTH ALABAMA

Spring 2026 Math Problem of the Week: Problem #6

Due noon Monday, February 23
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, February 23 or, if you cannot find me, you may leave it with Ms. Phifer in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #6: Compute the definite integral

$$\int_0^{\frac{\pi}{2}} \underbrace{\cos^2\left(\frac{\pi}{2}\cos^2\left(\frac{\pi}{2}\cos^2\left(\cdots\left(\frac{\pi}{2}\cos^2(x)\right)\cdots\right)\right)\right)}_{\text{2026 compositions}} \, dx$$